

Numerical investigation of topological states near magnetic structures on a superconductor

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Literature Review

Introduction. When magnetic impurities are in proximity to an s-wave superconductor, the magnetic scattering from the impurities breaks Cooper pairs creating in-gap energy states [1]. These so-called Yu-Shiba-Rusinov (YSR) states form close to the spatial position of the impurity. If the impurities are within sufficient proximity to each other, the confined states overlap and create degenerate energy bands in the superconducting gap. Sufficient magnetic scattering strength from the impurities will cause these bands to converge, closing the energy gap. If strong spin-orbit coupling is present, a topological phase transition occurs, which reopens the energy gap [1, 2, 3] and allows for the emergence of theoretical topological states at the edges of the chain called Majorana bound states (MBS).

These states are of great interest to topological quantum computing because MBS are non-Abelian anyons [4]. Non-Abelian anyons can be exchanged in a sequence non-trivially to produce a desired gate by a process called Braiding. Similar to fermions or bosons, multiple anyons can combine to form a composite anyon. A pair of spatially separated MBS form our qubit and the resulting states post-braiding are then combined via ‘fusion’ to create a parity protected qubit, which can then be measured and stored.

The process and the storage of the outcomes are considered fault-tolerant because they are protected topologically from local deformations that would usually lead to decoherence effects in a standard qubit. This is because the topological qubit depends on the global topological characteristics of the system, which makes it resilient to local disorder [5]. Despite this, TQC still has some undesired problems that could lead to decoherence. One of these is additional anyons appearing due to disorder that leads to undesired regions being in topological or trivial phases where they shouldn’t. Another big issue is quasi-particle poisoning. This is where electrons from an

external reservoir tunnel into the superconductor, causing the parity of the qubit to be undefined [6].

Key Studies. Although there is great hope in the field that these states can be utilised in this way in the future, there are no airtight experimental proofs that these states have been found. Thus, before TQC can be realised and the characteristics of MBS can be harnessed in a coherent fashion, they must be discovered with certainty. Attempts have been made to do this by looking for signatures of MBS in different systems that exhibit the properties of a topological superconductor.

A key signature of MBS is a localised zero bias conductance peak (ZBCP) in conductance-voltage measurements using scanning tunnelling microscopes or using quantum transport techniques [7] as shown in Fig. 1. To obtain ZBCP, it begins with attaching normal metals to both edges of the system. One acts as a source of electrons and the other acts as a drain that takes the electrons away. A gate voltage is applied that only allows the passage of the electrons over certain energy states. Thus, if there are any energy states at the threshold, a conductance will be measured. For an ideal superconductor, there is a band gap in the density of states, meaning there is zero conductance in the band gap. If $E = 0$ MBS states are present, a ZBCP appears because there are now energy states that allow the transport of electrons at the gate voltage tuned to zero energy, inducing a current and hence an inductance.

In reality, there are many types of disorder and contamination that can lead to in-gap energy states that are topologically trivial but give the appearance of MBS induced ZBCP. That being said, varying and fine tuning different parameters of the system like the magnetic field can filter out these types of causes. However, it is reasoned in Ref. [8] that it is likely that most ZBCP seen in recent studies are the result of strong random disorders that mimic all the characteristics of a non-trivial ZBCP. This means that although the

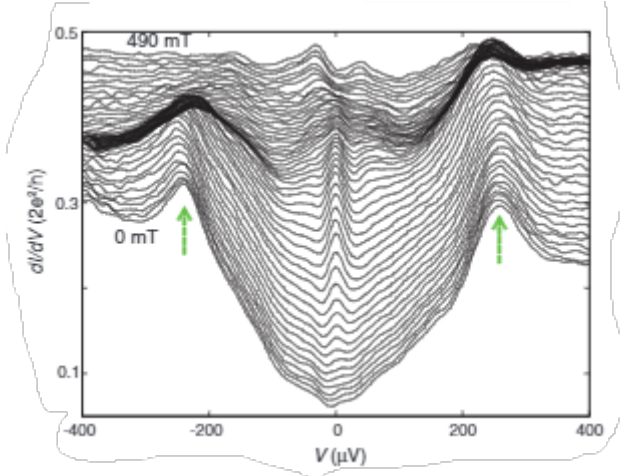


Figure 1: As seen in [10]. Conductance versus Voltage graph with varying magnetic field depicting ZBCP.

ZBCP is an inherent property of MBS, it cannot be reasoned that a ZBCP in a system believed to be topologically non-trivial is actually due to MBS.

Many approaches use these signatures to try to spot signs of MBS. One approach uses a semiconducting nanowire coupled to an s-wave superconductor in the presence of a magnetic field parallel to the nanowire (Zeeman field). With strong spin-orbit coupling, MBS can emerge if some of the characteristics of the system spatially vary [9]. Materials like InSb (Indium Antimonide) or InAs (Indium Arsenide) are chosen because they enable strong spin-orbit coupling and have large Lande g factors [10]. A large g factor is an important characteristic because of the Zeeman energy, $E_Z = g\mu_B B/2$ which is related to the topological phase condition in the following way:

$$E_Z = \sqrt{\Delta^2 + \mu^2} \quad (1)$$

where E_Z is the Zeeman energy, Δ is the gap function and μ is the chemical potential.

When the condition shown in Eq. (1) is met, a topological phase transition occurs, reopening the band gap, allowing for zero energy MBS's to emerge at both ends of the nanowire or on the domain walls. This condition shows that the system can be tuned by selecting the strength and which parameter to vary spatially.

An example of this type of set up is shown in Ref. [10]. In this experiment they use an InSb semiconductor nanowire, to enable strong spin-orbit coupling, connected to an Al (Aluminium) superconductor and a normal Au (Gold) electrode. In this experiment, they report the existence of

MBS due to a ZBCP that is not suppressed by changing the voltage or the magnetic field. This however could still be the result of sample impurities or strong disorder. Another issue is that the spatial origin of the ZBCP cannot be determined in this setup, making it harder to conclude whether they really are MBS [11].

Placing ferromagnetic or helical spin chains on the surface of a superconductor is another theoretical realisation of the system (see Fig. 2). This is because the modulated arrangement of the spins imitates the spin-orbit coupling and Zeeman field required to create a topological superconductor [12]. If the magnetic scattering of the impurities is sufficient, the gap closes and a topological phase transition occurs, reopening the energy gap. This allows for the emergence of MBS at both edges of the chain [13]. Experimentally, this type of system has some advantages over the semiconductor-superconductor system. For example, since the MBS are localised at the edges of the chains, a STM can be used to determine the spatial origin of the ZBCP associated with the topological phase transition. This is unlike the nanowire as previously mentioned. Although this is a positive, helical spin configurations are difficult to keep stable in practise and regress on an unstable trajectory towards ferromagnetic or anti-ferromagnetic arrangements [11].

Experiments based on this type of system are shown in Ref. [14, 12]. Both experiments report seeing signatures of MBS in the form of ZBCP. In Ref. [14] they place individual Mn (Manganese) impurities onto the surface of a Nb (Niobium) superconductor. This ensures that there are no defects in the system, which as previously mentioned, can be the cause of a topologically trivial ZBCP.

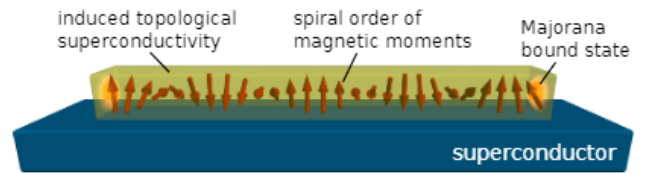


Figure 2: As seen in Ref. [13]. Helical spin-chains placed upon a superconductor that has undergone a topological phase transition, with Majorana bound states emerging at the ends of the chain.

Model. The model used in this project is equivalent to the non-self consistent approach used in Ref. [2]. The model uses the 2D BCS

tight-binding Hamiltonian for the superconductor shown in Eq. (2). Classical-spin magnetic impurities are placed upon the 2D square lattice substrate. Their interactions with the superconductor described by the impurity Hamiltonian shown in Eq. (3).

$$H_{BCS} = \sum_{\mathbf{r}} \left[-t \sum_{\mathbf{a}, \sigma} c_{\mathbf{r}+\mathbf{a}, \sigma}^{\dagger} c_{\mathbf{r}, \sigma} + (\Delta c_{\mathbf{r}, \uparrow}^{\dagger} c_{\mathbf{r}, \downarrow}^{\dagger} + \text{h.c.}) - \mu \sum_{\sigma} c_{\mathbf{r}, \sigma}^{\dagger} c_{\mathbf{r}, \sigma} \right] \quad (2)$$

where: $\mathbf{r}$ runs over all sites on the square lattice, $\mathbf{a}$ the nearest neighbour path vector, t the hopping amplitude, Δ the gap function, μ the chemical potential, σ the z-projection of the spin and $c_{\mathbf{r}, \sigma}$ the electron operator on the lattice sites.

$$H_{imp} = V_m \sum_{\mathbf{r} \in \mathcal{I}_m} \sum_{\sigma, \sigma'} \mathbf{S}_{\mathbf{r}} \cdot \boldsymbol{\sigma}_{\sigma, \sigma'} c_{\mathbf{r}, \sigma}^{\dagger} c_{\mathbf{r}, \sigma'} + V_s \sum_{\mathbf{r} \in \mathcal{I}_s} \sum_{\sigma} c_{\mathbf{r}, \sigma}^{\dagger} c_{\mathbf{r}, \sigma} \quad (3)$$

where: S_r the spin of an impurity located at $\mathbf{r} \in \mathcal{I}_m$, $\boldsymbol{\sigma}$ the set of Pauli matrices, V_m the magnetic scattering strength, V_s the scalar impurity scattering strength and $\mathcal{I}_m, \mathcal{I}_s$ are the locations of sites where magnetic impurities and scalar impurities are located, respectively.

In this approach the gap-function Δ is constant and specified manually. It's value is calculated far from the region of interest by a self-consistent model using fixed point iteration. In contrast to our constant Δ , the self consistent approach produces a small dip in the density of states at the sites of impurities since they scatter electrons outward. This method is applicable because there is negligible qualitative difference in the band structure around $E=0$ when you use a constant pre-determined gap function [2]. With this comes the assumption that the attractive electron interaction V is tuned to the corresponding value using $\Delta = V \langle c^{\dagger} c^{\dagger} \rangle$.

In this model, we neglect the following: spin-orbit interaction, Dzyaloshinskii-Moria interaction and any non s-wave singlet pairing. Spin-orbit coupling is a relativistic effect where an electron within an orbital is travelling at relativistic speeds inside an electric field, producing a magnetic field in its frame of reference, which interacts with the spin of the electron itself. This interaction can be neglected because the spin arrangement of the impurities on the

2D superconductor substrate imitates the interaction, meaning we can still capture the physics present. The Dzyaloshinskii-Moria interaction can be neglected for a similar reason because it resembles spin-orbit coupling but on a lattice. Non s-wave singlet pairing can also be neglected since in this system the component electrons that combine to create the cooper pairs are of equal but opposite-sign momenta, meaning they have zero C-O-M momentum. Therefore, under particle exchange, the spatial wave function is symmetric which means the spins of the states must be paired in the anti-symmetric singlet state such that the cooper pair obeys the Pauli exclusion principle.

Developments. This project will begin with the testing of the validity of the program against a previous MPhys student's analytical derivation of the band structure of a ring of impurities (See Fig. 3A) placed upon a s-wave superconductor [15]. In this project, the Green's function for the ring structure was derived analytically. From this, the density of states and the topological Hamiltonian's were calculated. It was concluded that topological phase transitions occurred and that they were strongly affected by ring arrangement. After verifying these results, an incision will be made into the ring structure, leaving two edges as shown in Fig. 3B. The band structure near $E=0$ will then be computed to see whether MBS emerge as a consequence. Following this, more

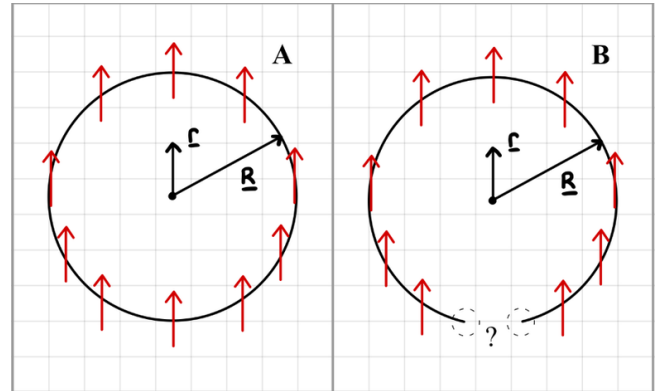


Figure 3: (A) Adapted figure from Ref. [15] Ring arrangement of ferromagnetic impurities on a 2D superconductor lattice substrate. (B) Ring arrangement of magnetic impurities with an incision into the structure, creating a gap with two edges.

complex structures with vertices and holes will be computed to determine whether they display any interesting topological properties and if they allow

MBS to materialise.

The influence of finite size effects due to the overlapping of the limited wave functions (and degrees of freedom) available in the system will need to be considered. This can be determined over iterations of changing the dimensions and boundary conditions (soft, periodic, open) of the system. From this, it is hoped that these effects can be mitigated and tuned out. An investigation into how the properties of the system change with the different boundary conditions, once the ideal parameters of the system have been found, could be interesting as if there are no effects then the most computationally efficient boundary condition could be used hence forth. It could also be intriguing to see if the spatial proximity of edges to one another (Fig. 3B) affects the topological phase transition in a similar way to short-spin chains mentioned in Ref. [12] where the topological phase transition was suppressed due to the edge states being coupled together over the short distance.

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